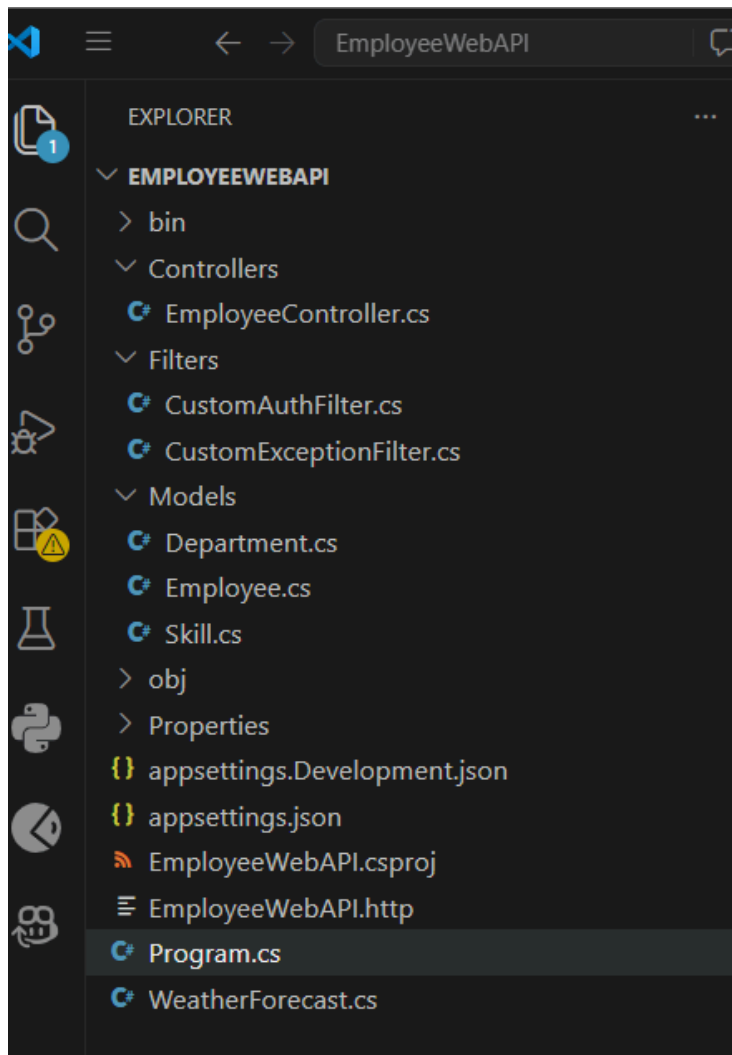


Lab-3: Web API

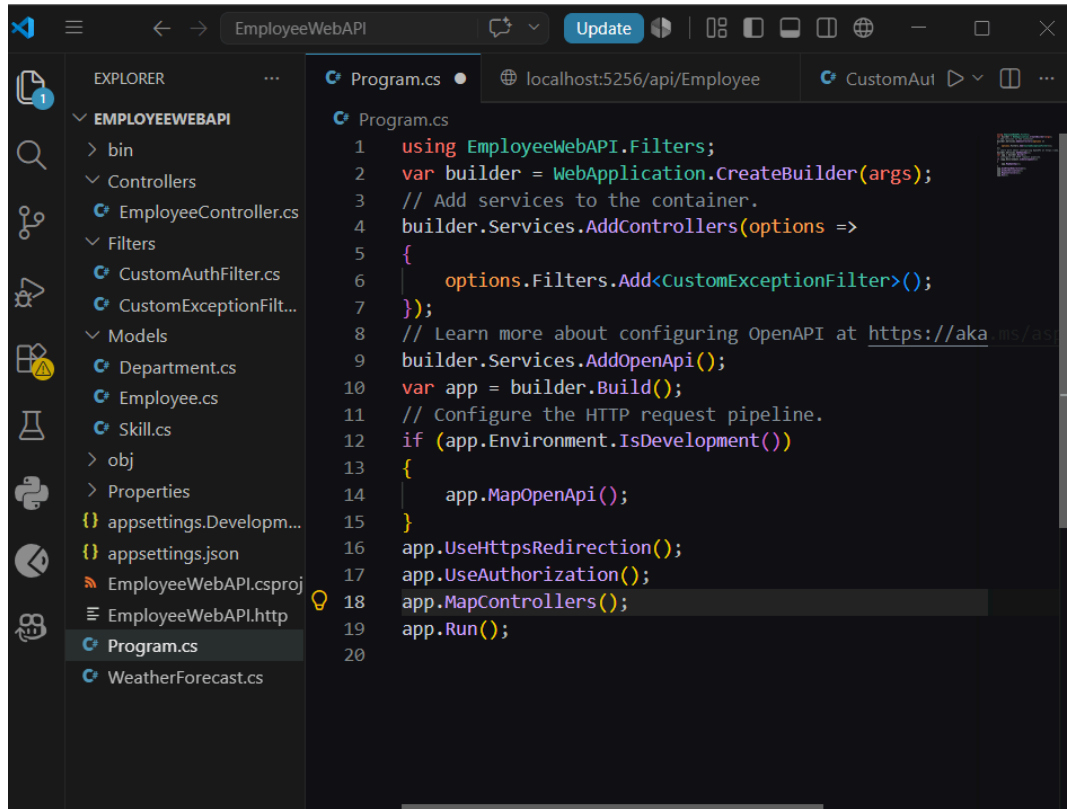
Aim

To develop an ASP.NET Core Web API application that demonstrates employee management, custom authorization filtering, and exception handling using filters.

Folder Structure:

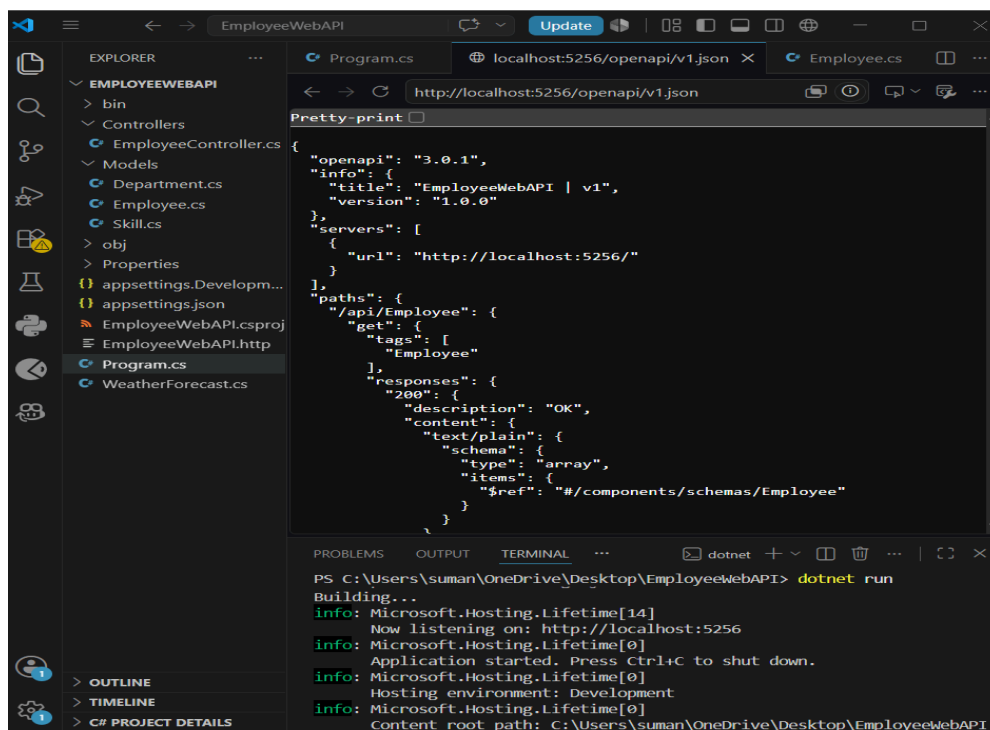


Program.cs:



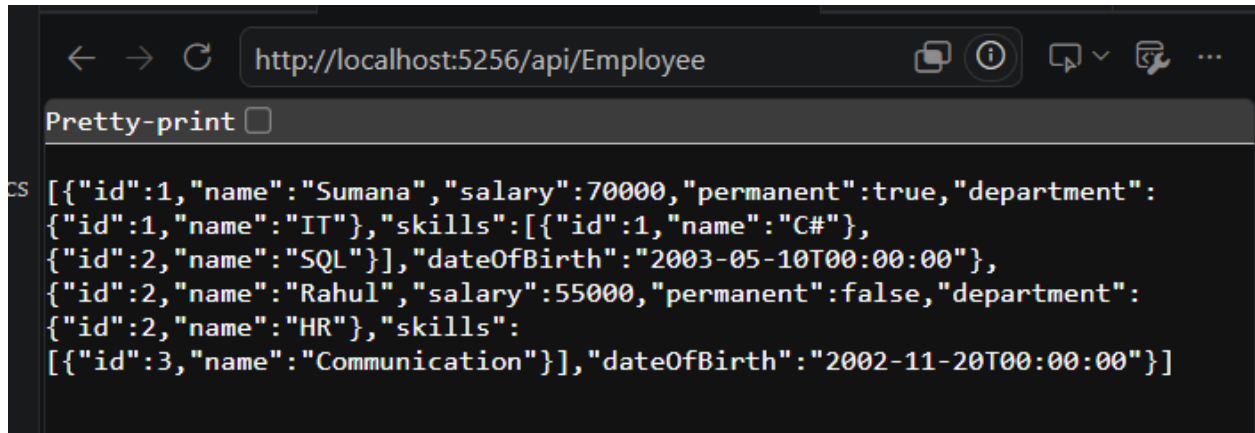
```
1 using EmployeeWebAPI.Filters;
2 var builder = WebApplication.CreateBuilder(args);
3 // Add services to the container.
4 builder.Services.AddControllers(options =>
5 {
6     options.Filters.Add<CustomExceptionFilter>();
7 });
8 // Learn more about configuring OpenAPI at https://aka.ms/aspnetcore/openapi
9 builder.Services.AddOpenApi();
10 var app = builder.Build();
11 // Configure the HTTP request pipeline.
12 if (app.Environment.IsDevelopment())
13 {
14     app.MapOpenApi();
15 }
16 app.UseHttpsRedirection();
17 app.UseAuthorization();
18 app.MapControllers();
19 app.Run();
20
```

Output:



```
{
  "openapi": "3.0.1",
  "info": {
    "title": "EmployeeWebAPI | v1",
    "version": "1.0.0"
  },
  "servers": [
    {
      "url": "http://localhost:5256/"
    }
  ],
  "paths": {
    "/api/Employee": {
      "get": {
        "tags": [
          "Employee"
        ],
        "responses": {
          "200": {
            "description": "OK",
            "content": {
              "text/plain": {
                "schema": {
                  "type": "array",
                  "items": {
                    "$ref": "#/components/schemas/Employee"
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

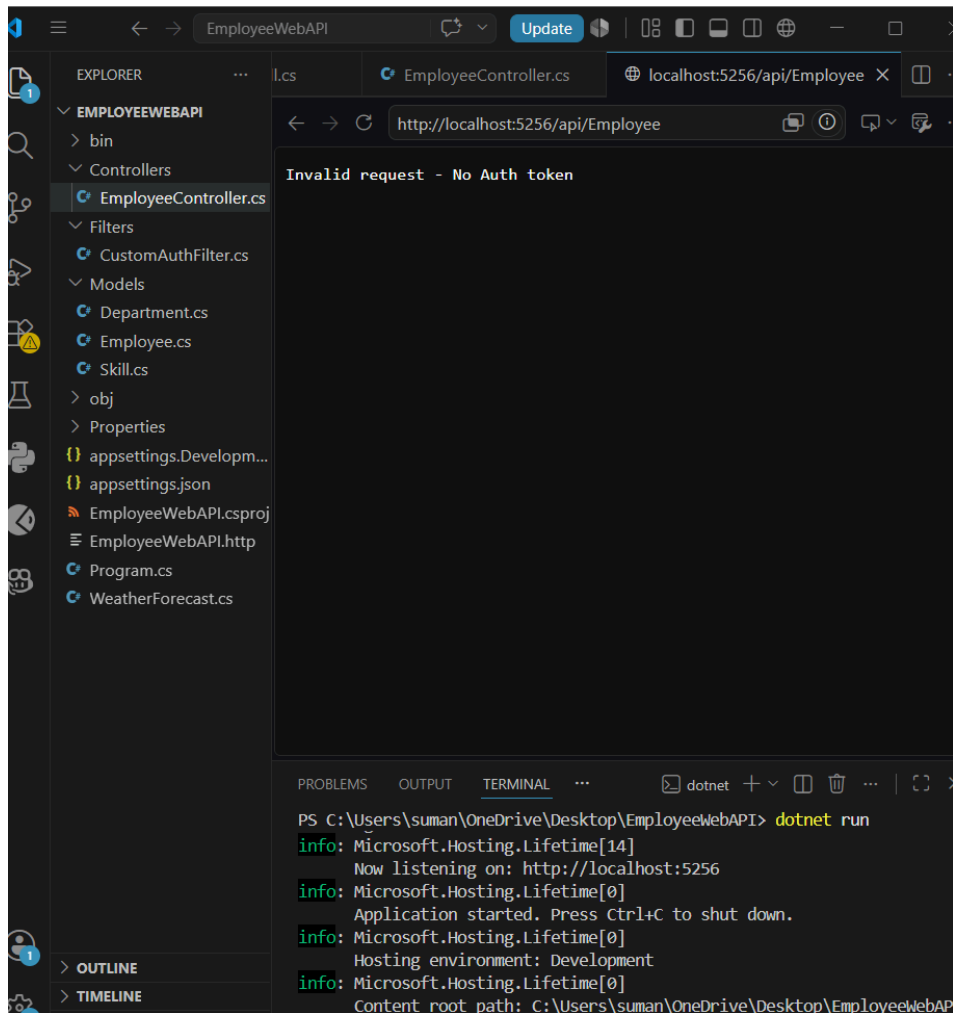
```
PS C:\Users\suman\OneDrive\Desktop\EmployeeWebAPI> dotnet run
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5256
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\suman\OneDrive\Desktop\EmployeeWebAPI
```



```
http://localhost:5256/api/Employee

Pretty-print ☐

[{"id":1,"name":"Sumana","salary":70000,"permanent":true,"department":{"id":1,"name":"IT"},"skills":[{"id":1,"name":"C#"}, {"id":2,"name":"SQL"}],"dateOfBirth":"2003-05-10T00:00:00"}, {"id":2,"name":"Rahul","salary":55000,"permanent":false,"department":{"id":2,"name":"HR"},"skills":[{"id":3,"name":"Communication"}],"dateOfBirth":"2002-11-20T00:00:00"}]
```



Result

Successfully created and tested an ASP.NET Core Web API with custom model classes, implemented GET, POST, and PUT operations, applied a custom authorization filter, and handled exceptions using a custom exception filter.